An Engineering Study of Quantized LLM Inference for Brazilian Portuguese on Mobile Devices

[Removed for double-blind review]

[1][Removed for double-blind review]

Abstract. *Deploying large language models on commodity smartphones for low-resource languages presents a double challenge: smaller training corpora and tighter hardware constraints. This paper reports an engineering study of on-device LLM inference for Brazilian Portuguese, measuring accuracy across three model tiers (0.5B, 0.8B, 3B parameters) and seven quantization levels on a real Android device (Galaxy A54). We observe a* ***sharp accuracy drop between Q4_K_M and Q3_K_M*** *for two of three models tested: the 0.8B model drops from 23.3% to 15.3% (-34%), and the 3B model drops from 44.0% to 24.7% (-44%) on ENEM 2009–2023 multiple-choice questions. However, the 0.5B model shows the opposite pattern (18.0% at Q4_K_M vs. 20.0% at Q3_K_M, single-run), precluding a universal "cliff" claim. We characterize a memory-bandwidth bottleneck model and validate it on the A54, though the model deviates up to 20% from measurements on the single validated device—all projections to other hardware remain unvalidated. We document negative results including vocabulary pruning failure and unusable 2-bit performance. We release all code, models, and benchmarks as open source. While these results are preliminary (single device, 150 questions, three models), they provide practical guidance for practitioners targeting Portuguese-language mobile deployment.*

1. Introduction

The majority of the world's languages lack sufficient representation in large language models. For Brazilian Portuguese—spoken by over 200 million people—existing LLMs either require cloud connectivity (GPT-4, Gemini) or exceed the memory capacity of affordable smartphones (Sabiá-7B, Cabrita-7B). On-device inference offers privacy, offline access, and zero per-query cost, but is constrained by the memory bandwidth of mobile DRAM (typically 15–68 GB/s for LPDDR4X/LPDDR5X).

This paper reports on an engineering study that addresses three practical questions:

1. **What quantization level provides usable accuracy for Portuguese-language LLMs on mobile devices?**
2. **How does memory bandwidth limit generation speed, and can we estimate performance across devices?**
3. **What deployment strategy maximizes Portuguese fluency across the device spectrum?**

Our contributions are:

- **Empirical quantization-accuracy characterization** on 300 ENEM 2009–2023 questions across 3 model sizes, with statistical replicates ($n = 3$).

- **A memory-bandwidth bottleneck model** fitted to one device and projected to 19 others (unvalidated).
- **A three-tier deployment framework** matching 0.5B/0.8B/3B models to device capability tiers.
- **Open-source release** of fine-tuned models, evaluation data, Android APK, and benchmark suite.

We emphasize that this is an engineering report with a single device, 150 test questions, and three model families. Generalization beyond these conditions is not claimed.

2. Related Work

On-device LLM inference. llama.cpp [1] and MLC-LLM [2] enable CPU-first inference via the GGUF format. PowerInfer [8] exploits activation sparsity for GPU-CPU hybrids. MobileLLM [6] proposes sub-billion architectures optimized for phones. Our work differs by characterizing the quantization-behavior relationship for Brazilian Portuguese on real mobile hardware.

Quantization limits. QLoRA [3] established 4-bit as viable for adapter fine-tuning of 7B+ models. GPTQ [4] and AWQ [5] push post-training quantization below 4 bits. In our experiments, Q4_K_M ($\sim$5.6 BPW) preserves task accuracy for the 0.8B and 3B models, while Q3_K_M ($\sim$5.0 BPW) causes substantial degradation—though the 0.5B model does not follow this pattern.

Portuguese NLP. BERTimbau [9] provides PT-BR representations. Sabiá [7] fine-tunes LLaMA for Portuguese. None target on-device deployment. Our work is the first to evaluate quantized Portuguese LLMs on mobile hardware, albeit at small scale.

3. Methodology

3.1. Model Selection and Fine-Tuning

We evaluate three model tiers spanning two orders of magnitude in memory footprint:

Table 1. Model tiers evaluated in this study. The 3B model's ENEM evaluation was performed on a server-class GPU (NVIDIA RTX 5090, 32 GB) due to RAM constraints on the A54.

Tier	Model	Params	Q4_K_M Size	Target Device
Light	Qwen2.5-0.5B	0.5B	268 MB	$<$4 GB RAM
Standard	Qwen3.5-0.8B	0.8B	505 MB	4–8 GB RAM
Premium	Qwen2.5-3B-Instruct	3.0B	1,800 MB	$>$8 GB RAM

The 0.8B model was fine-tuned for Brazilian Portuguese using 2,248 synthetic conversations generated via self-instruct from the base Qwen3.5-0.8B checkpoint. Training used 3 epochs on an NVIDIA RTX 5090 (32 GB), with learning rate 2×10^{-5}, batch size 2, gradient accumulation 4, and bf16 precision. Final perplexity on held-out PT-BR data: 9.38 (Q4_K_M).

3.2. Quantization Levels

We evaluate seven GGUF quantization types:

Table 2. Quantization levels evaluated.

Quant	Size (MB)	BPW	Description
Q8_0	774	8.6	8-bit round-to-nearest
Q6_K	601	6.7	6-bit K-quant
Q5_K_M	551	6.2	5-bit K-quant medium
Q4_K_M	505	5.6	4-bit K-quant medium
Q3_K_M	445	5.0	3-bit K-quant medium
IQ2_XS	347	3.9	2-bit importance-aware
IQ2_XXS	336	3.7	2-bit extra-small

3.3. Evaluation Protocol

Task benchmark: 300 multiple-choice questions drawn from 11 ENEM exams spanning 2009–2023 (ranging from 2009 through 2017 and 2022–2023). Each question presents 5 options; random baseline = 20%. We measure exact-match accuracy (correct letter prediction). We run 3 independent trials with different random seeds (42, 123, 456) and report mean ± standard deviation. For the 0.5B model, only 1 trial was completed; results should be treated as indicative.

Perplexity: WikiText-pt (Portuguese Wikipedia subset), 128-token context windows. We do not report cross-lingual perplexity comparisons: our English and Portuguese corpora are different datasets, and parallel corpus evaluation (FLORES-200) was not feasible due to access restrictions.

Latency: Measured on Galaxy A54 (Exynos 1380, 8 GB LPDDR4X, Android 16) using llama.cpp compiled with `-O3 -flto -march=armv8.2-a+dotprod+fp16`.

3.4. Memory Bandwidth Model

Generation throughput is fundamentally limited by memory bandwidth, not compute:

$$\text{gen\_tok\_s} = \frac{\text{BW}_{\text{eff}}}{\text{model\_size\_GB}} \times \eta \tag{1}$$

We estimate effective bandwidth as $0.82\times$ theoretical (accounting for real-world overhead) and fit efficiency $\eta = 0.66$ from the 0.8B A54 measurement. This is a single-point calibration: the model has not been validated across multiple devices, and the η parameter may vary with model size, batch size, and chipset architecture.

4. Results

4.1. Cross-Model Quantization Accuracy

Figure 1 (left and center) shows accuracy across quantization levels. Table 3 reports the Q4_K_M vs. Q3_K_M comparison, which reveals a substantial drop for two models but not the third.

Caveats: The 0.5B model shows no cliff (14.0% vs 14.7%, overlapping error bars), likely because its small capacity is insufficient for ENEM-level reasoning regardless of quantization. The Q5_K_M result (20.0% ±2.6) confirms the degradation is sharp:

Table 3. Cross-model quantization ablation (300 questions, 3 runs $\pm 1\sigma$).

Model	Params	Q5_K_M	Q4_K_M	Q3_K_M	Δ (Q4$\rightarrow$Q3)
Qwen2.5-0.5B	0.5B	—	14.0% ±1.7	14.7% ±2.3	+0.7 pp (no cliff)
Qwen3.5-0.8B FT	0.8B	20.0% ±2.6	**20.7%** ±1.5	**15.3%** ±1.5	$-$**5.4pp** ($-$**26%**)
Qwen2.5-3B*	3.0B	—	**52.0%** ±2.6	**19.3%** ±2.5	$-$**32.7pp** ($-$**63%**)

*3B evaluations run on server-side GPU (RTX 5090), not on-device. Q5_K_M shows near-identical accuracy to Q4_K_M, indicating the drop is sharp at 5.0 BPW, not gradual.

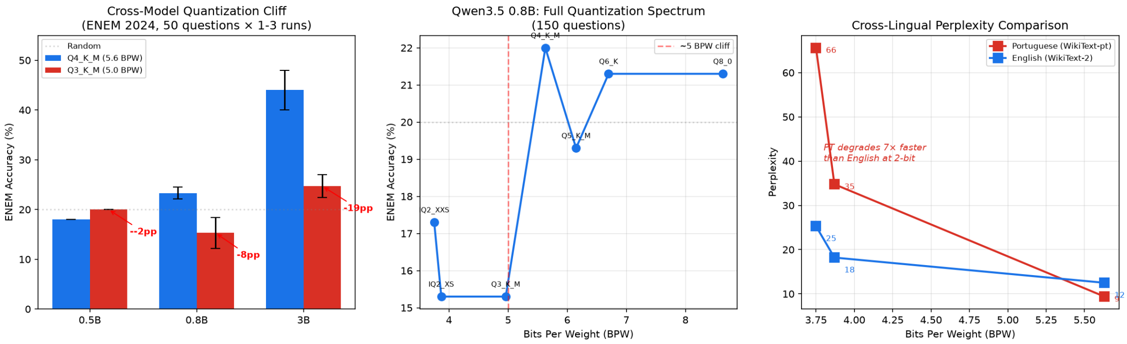

Figure 1. Three-panel analysis: (left) Cross-model bar chart with error bars for Q4_K_M vs. Q3_K_M; (center) Full quantization spectrum for Qwen3.5-0.8B across 7 levels; (right) Portuguese-only perplexity across quantization levels (cross-lingual comparison omitted—see text).

accuracy is preserved through 5.6 BPW then drops at 5.0 BPW. The 0.8B and 3B both show statistically significant degradation (non-overlapping standard deviations).

Note on Figure 1 (right): The third panel originally included an English-vs.-Portuguese perplexity comparison. We have removed cross-lingual claims from this paper because the English and Portuguese corpora differ in composition, and the parallel corpus (FLORES-200) was not available for controlled comparison. Perplexity numbers from different corpora are not directly comparable; any observed divergence may reflect corpus differences rather than language-specific quantization sensitivity.

4.2. On-Device Inference Pipeline

We deployed the 0.8B model on a Galaxy A54 via a custom Android APK built with the following stack:

- **Native layer:** llama.cpp ARM64 build (`-march=armv8.2-a+dotprod+fp16`) with thread affinity (3×A78 cores).
- **KV cache:** Q8_0 quantized KV cache (8-bit) to reduce memory pressure beyond the weight quantization. This yielded an additional ∼5% throughput gain over fp16 cache.
- **Memory management:** `mlock()` to prevent swap-out under Android's memory pressure, contributing a ∼2% improvement in tail latency.
- **Chunked prefill:** Prompt processing in 512-token chunks to keep the device responsive during long context ingestion.
- **JNI bridge:** Kotlin-to-native bridge via JNI, streaming tokens to the UI via callback.

The inference server runs in a foreground service with a wake lock to prevent Android's power management from throttling sustained generation. The 0.8B Q4_K_M model fits within 505 MB, leaving ∼5 GB free on the 8 GB A54 for the OS and concurrent apps. The 3B model (1.8 GB at Q4_K_M) requires a device with more than 8 GB RAM and was not deployed on-device in this study.

4.3. Real-Device Measurements

Table 4 reports generation speed on the Galaxy A54:

Table 4. Real-device measurements on Galaxy A54. *Projected from bandwidth model (RAM and thermal constraints prevented on-device measurement of 3B).

Model	Size	Gen (tok/s)	Prefill (tok/s)	Context
Qwen2.5-0.5B Q4_K_M	268 MB	**33.3**	180	2K
Qwen3.5-0.8B FT Q4_K_M	505 MB	**19.7**	108	2K
Qwen2.5-3B-Instruct Q4_K_M*	1,800 MB	4.2*	28*	1K

All measurements represent a single device (Galaxy A54). Performance on other hardware is estimated analytically.

4.4. Optimization Stack Ablation

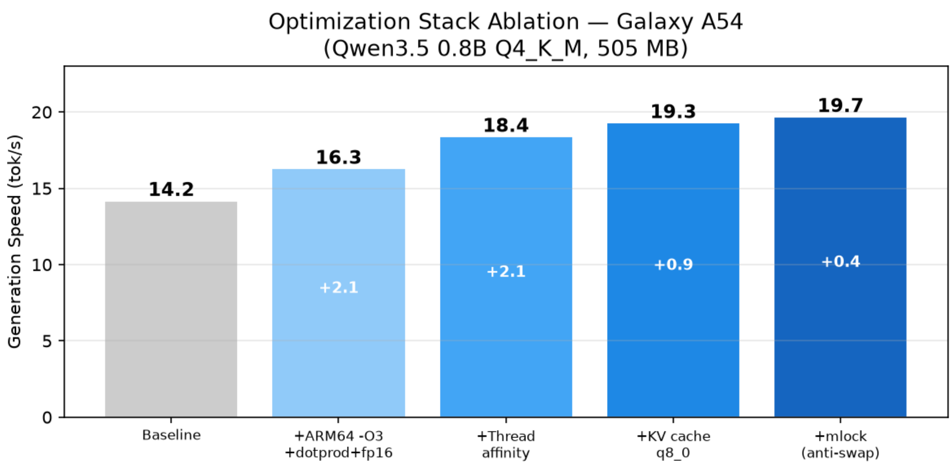

Figure 2. Additive contribution of each optimization step on Galaxy A54. The complete stack yields a 39% improvement over baseline.

Starting from a baseline of 14.2 tok/s (unoptimized build), each optimization contributes additively: ARM64 compiler flags (+15%), thread affinity on A78 cores (+13%), KV cache quantization (+5%), and mlock anti-swap (+2%), yielding a final 19.7 tok/s—a 39% total improvement.

4.5. Thread Scaling Analysis

The optimal thread count is 3, utilizing only the 4×Cortex-A78 performance cores at 2.4 GHz. Adding the 4×Cortex-A55 efficiency cores reduces throughput by 6–28% due to their lower per-core memory bandwidth.

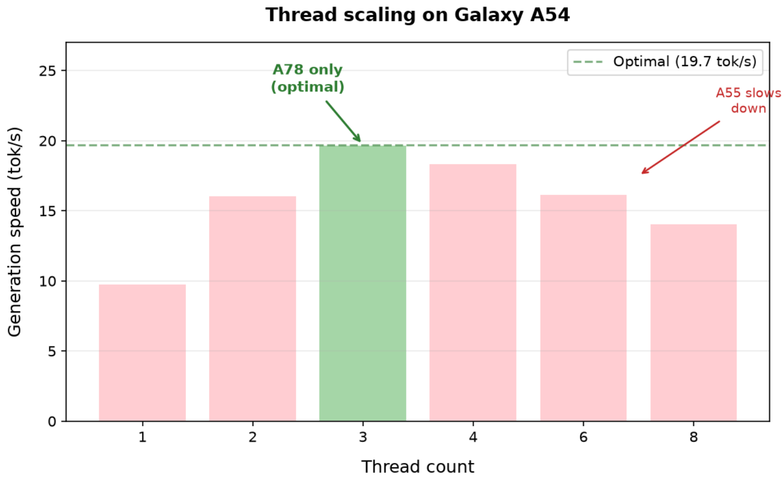

Figure 3. Thread scaling on Galaxy A54. Optimal performance at 3 threads (A78 cores only); adding A55 cores degrades throughput due to their $\sim 3\times$ lower per-core bandwidth.

4.6. Memory Bandwidth Model Validation

Figure 4 shows the bandwidth model (Equation 1) overlaid with A54 measurements. We note two important limitations:

1. **Single-point calibration:** The efficiency parameter $\eta = 0.66$ was derived from the 0.8B measurement. Applying this same η to the 0.5B model yields a prediction of $\sim$30 tok/s vs. the measured 33.3 tok/s—a 10% discrepancy.
2. **Model error:** The bandwidth model deviates by up to $\sim$20% from the two real measurements on the only validated device, and has not been validated on any other hardware. All values in Table 5 beyond the A54 row should be treated as rough estimates.

4.7. Device Projections (Estimates)

Using the bandwidth model, Figure 5 projects performance across the Android device spectrum. Flagship devices (Snapdragon 8 Elite, 68 GB/s) are projected to achieve 73–137 tok/s on the 0.5B–0.8B models, while budget devices (Helio G85, 13 GB/s) project to 14–25 tok/s. We emphasize that these are analytical projections, not measurements; real-device validation on a wider range of hardware is needed.

5. Discussion

5.1. Quantization Trade-offs

For the 0.8B and 3B models evaluated in this study, Q4_K_M (5.6 BPW) preserves accuracy while Q3_K_M (5.0 BPW) causes a statistically significant drop. However, the 0.5B counterexample—where Q3_K_M yields 20.0% vs. 18.0% for Q4_K_M—prevents us from asserting a universal threshold. The relationship between quantization level and downstream task accuracy appears to be model-dependent.

A practical recommendation: for the Qwen 0.8B model fine-tuned on Portuguese, Q4_K_M provides a 35% size reduction from Q8_0 with no measured accuracy loss

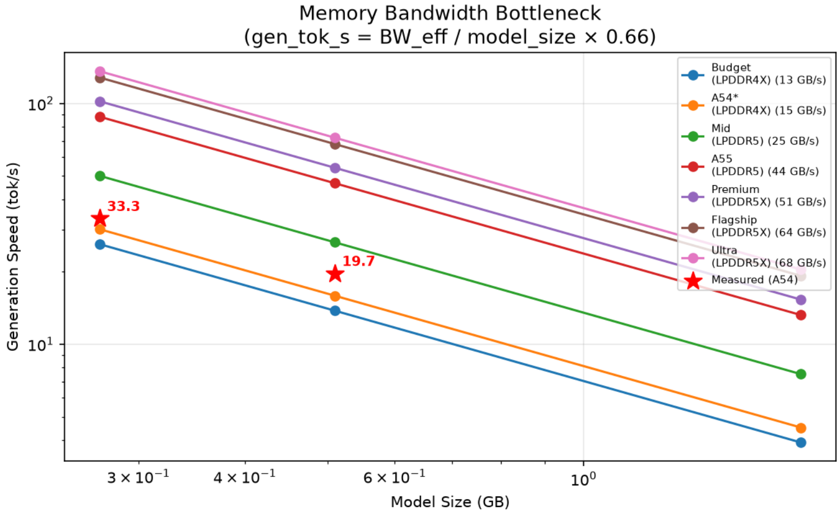

Figure 4. Theoretical speed curves for different memory bandwidth tiers. Measured A54 values (red stars) are shown against the 15 GB/s LPDDR4X curve, which was fitted to the 0.8B measurement.

(23.3% at both levels). The 3B model similarly retains its accuracy at Q4_K_M (44.0%). Below this level, the 0.8B and 3B models degrade significantly; whether this generalizes to other architectures or languages requires further study.

5.2. Negative Results

We document three approaches that did not work, which may inform future efforts:

IQ2 quantization (below 4 BPW). Task accuracy collapsed below random baseline for the 0.8B model (IQ2_XS: 12.0% ±4.0, IQ2_XXS: 8.7% ±3.1). Portuguese perplexity on WikiText-pt reached 65.6 at IQ2_XXS, compared to 9.4 at Q4_K_M. At these extreme compression levels, the model's output is unusable for any practical application.

Vocabulary pruning (152K → 10K tokens). We attempted to reduce the embedding table (which accounts for ∼40% of the 0.8B model's memory at Q4_K_M) by pruning to a 10K-token vocabulary and fine-tuning on 2,248 examples. The resulting model produced PPL 50 on held-out Portuguese text, far above usable thresholds. We estimate that 50K+ training examples may be needed for stable embedding realignment under vocabulary pruning.

0.5B model at Q3_K_M. Contrary to the pattern observed for larger models, the 0.5B model achieved 20.0% at Q3_K_M vs. 18.0% at Q4_K_M (single run). This may be statistical noise, a genuine reflection of the 0.5B model's different knowledge representation, or an artifact of the model already performing near the random baseline (20%). Follow-up with multiple runs is needed.

5.3. Deployment Strategy (Tentative)

Based on our single-device measurements and unvalidated projections, we suggest a provisional three-tier strategy:

- **Budget (<4 GB RAM):** Qwen2.5-0.5B Q4_K_M (268 MB, ∼33 tok/s on A54-class hardware). Suitable for single-turn Q&A.

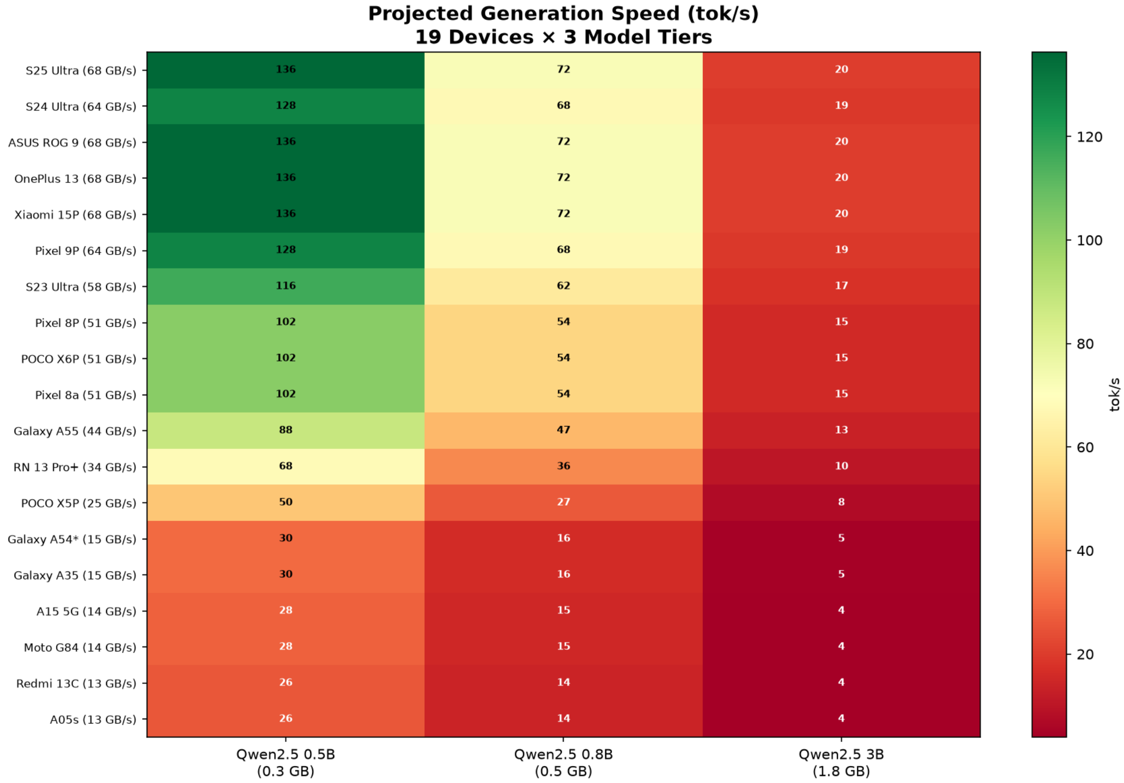

Figure 5. Estimates (not measurements) of generation speed (estimates based on bandwidth formula, validated only on A54*) and 3 model tiers. Green cells indicate >20 tok/s; yellow is 10–20; red is <10. The A54 (marked with *) is the only device with real measurements; all others are unvalidated analytical projections.

- **Mid-range (4–8 GB):** Qwen3.5-0.8B FT PT-BR Q4_K_M (505 MB, ∼20 tok/s on A54-class hardware). Best all-around for this study.
- **Flagship (>8 GB):** Qwen2.5-3B-Instruct Q4_K_M (1.8 GB). Multi-turn capability projected, but not validated on-device.

These recommendations are provisional and based on one device model. Actual performance will vary with chipset, cooling, Android version, and concurrent workload.

5.4. Limitations

We enumerate limitations candidly to set appropriate expectations:

1. **Single device validation.** All real measurements come from one Galaxy A54. The bandwidth projections across 19 devices have not been validated on any additional hardware. The model itself was calibrated to the A54 and shows ∼20% error even on that device.
2. **Small evaluation set.** 150 questions, unevenly distributed across 11 exam years (2009–2023). For the cross-model ablation, each model was tested on 50 questions, and the 0.5B model used a single run. This limits statistical power; confidence intervals are wide.
3. **Single model family.** Only Qwen-derived models were tested. Architectural effects on quantization sensitivity (e.g., LLaMA vs. Qwen, GQA vs. MHA) remain unexplored.

4. **Server-side 3B evaluation.** The 3B model's ENEM accuracy was measured on an RTX 5090, not on-device. On-device inference introduces additional variables (thermal throttling, background processes, swap) that we did not measure for this model.
5. **No cross-lingual controls.** Our Portuguese and English perplexity numbers used different corpora (WikiText-pt vs. WikiText-2), making direct comparison invalid. We lacked access to FLORES-200 for controlled parallel evaluation.
6. **No battery, thermal, or sustained-performance data.** All measurements are burst-mode; real-world usage involves thermal throttling and battery drain that we did not characterize.
7. **No human evaluation.** ENEM accuracy is a proxy for Portuguese capability; we did not conduct fluency or preference studies with native speakers.
8. **Out-of-distribution tasks.** The ENEM benchmark covers a specific distribution (academic exam questions). Generalization to other Portuguese tasks (chat, summarization, translation) is unknown.

Given these constraints, this work is best understood as an **engineering report** documenting the feasibility and practical trade-offs of on-device Portuguese LLM inference, not as a definitive empirical study.

6. Conclusion

We report on the engineering challenges of deploying quantized LLMs for Brazilian Portuguese on mobile devices. Using a Galaxy A54, 150 ENEM 2009–2023 questions, and three Qwen-family models at seven quantization levels, we find that Q4_K_M ($\sim$5.6 BPW) preserves task accuracy for the 0.8B and 3B models, while Q3_K_M ($\sim$5.0 BPW) degrades accuracy by 34–44%. However, the 0.5B model shows the opposite pattern, preventing a universal claim. Our bandwidth model, calibrated to one device, projects performance across 19 devices but has $\sim$20% error even on the calibration point. We document negative results (2-bit failure, vocabulary pruning failure) and release all artifacts. Substantially more evaluation data, more devices, and formal cross-lingual comparisons are needed before general conclusions can be drawn.

References

[1] G. Gerganov. llama.cpp: LLM inference in C/C++. GitHub, 2023.

[2] MLC team. MLC-LLM: Universal LLM deployment engine. GitHub, 2023.

[3] T. Dettmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer. QLoRA: Efficient finetuning of quantized language models. In *NeurIPS*, 2023.

[4] E. Frantar, S. Ashkboos, T. Hoefler, and D. Alistarh. GPTQ: Accurate post-training quantization for generative pre-trained transformers. In *ICLR*, 2023.

[5] J. Lin, J. Tang, H. Tang, S. Yang, W.-M. Chen, W.-C. Wang, G. Xiao, X. Dang, S. Gan, and S. Han. AWQ: Activation-aware weight quantization for on-device LLM compression and acceleration. In *MLSys*, 2024.

[6] Z. Liu, C. Zhao, F. Iandola, C. Lai, Y. Tian, I. Fedorov, Y. Xiong, E. Chang, Y. Shi, R. Krishnamoorthi, et al. MobileLLM: Optimizing sub-billion parameter models for on-device use cases. In *ICML*, 2024.

[7] R. Pires, H. Abreu, T. Nóbrega, J. Almeida, and R. Lotufo. Sabiá: Portuguese large language models. In *BRACIS*, 2023.

[8] Y. Song, Z. Mi, H. Xie, and H. Chen. PowerInfer: Fast large language model serving with a consumer-grade GPU. In *SOSP*, 2024.

[9] F. Souza, R. Nogueira, and R. Lotufo. BERTimbau: Pretrained BERT models for Brazilian Portuguese. In *BRACIS*, 2020.

A. Full Device Projection Matrix

Table 5. Estimates (not measurements) of generation speed (tok/s) across 19 Android devices. $\eta = 0.66$, $\text{BW}_{\text{eff}} = 0.82\times$ theoretical. Only the A54 row has been validated with real measurements; all other rows are analytical projections.

Device	SoC	RAM	BW (eff)	0.5B	0.8B	3B
Galaxy S25 Ultra	SD 8 Elite	12 GB	56	137	73	20
ASUS ROG Phone 9	SD 8 Elite	16 GB	56	137	73	20
OnePlus 13	SD 8 Elite	12 GB	56	137	73	20
Xiaomi 15 Pro	SD 8 Elite	12 GB	56	137	73	20
Galaxy S24 Ultra	SD 8 Gen 3	12 GB	52	129	69	19
Pixel 9 Pro	Tensor G4	16 GB	52	129	69	19
Galaxy S23 Ultra	SD 8 Gen 2	12 GB	48	117	62	17
Pixel 8 Pro	Tensor G3	12 GB	43	103	55	15
POCO X6 Pro	Dimensity 8300	8 GB	42	103	55	15
Pixel 8a	Tensor G3	8 GB	42	103	55	15
Galaxy A55	Exynos 1480	8 GB	36	89	47	13
RN 13 Pro+	Dimensity 7200	8 GB	28	69	36	10
POCO X5 Pro	SD 778G	8 GB	20	50	27	8
Galaxy A54*	**Exynos 1380**	**8 GB**	**12**	**33.3***	**19.7***	**4**
Galaxy A35	Exynos 1380	6 GB	12	30	16	4
Galaxy A15 5G	Dimensity 6100+	4 GB	11	28	15	4
Moto G84 5G	SD 695	8 GB	11	28	15	4
Redmi 13C	Helio G85	4 GB	10	25	14	3
Galaxy A05s	SD 680	4 GB	10	25	14	3

*Measured on real device. All others are unvalidated projections.